

# National University of Computer and Emerging Sciences, Lahore Campus



|             |                                    |              |             |
|-------------|------------------------------------|--------------|-------------|
| Course:     | Parallel and Distributed Computing | Course Code: | CS-3006     |
| Program:    | BSCS & BSDS                        | Semester:    | Spring 2024 |
| Duration:   | 20 Minutes                         | Total Marks: | 16          |
| Paper Date: | 14-May-2024                        | Weight       | 3.34%       |
| Type:       | Announced                          | Page(s):     | 2           |
| Exam:       | Quiz 4B Solution                   | Section:     |             |

Name & Roll No:

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## Question # 1:

[8 + 2 marks, CLO # 2]

a) Write the output for the following piece of code assuming that there are 8 MPI processes & no errors.

```
#include <mpi.h>
#include <stdio.h>
#include <cmath>

int main(int argc, char** argv) {
    MPI_Init(NULL, NULL);
    MPI_Status status;
    int p, recVal, my_rank, neighbor;
    MPI_Comm_size(MPI_COMM_WORLD, &p);
    MPI_Comm_rank(MPI_COMM_WORLD, &my_rank);

    int* d_val = new int[p];
    int myVal = pow(2, my_rank+1); // 2^(my_rank+1)
    int sTag = my_rank;
    if (my_rank % 2 == 0) neighbor = (my_rank + 1) % p;
    else neighbor = (my_rank - 1) % p;
    MPI_Sendrecv(&myVal, 1, MPI_INT, neighbor, sTag, &recVal, 1, MPI_INT, neighbor, neighbor,
    MPI_COMM_WORLD, &status);
    int diff = abs(myVal - recVal); // absolute value
    if (my_rank > 0 && my_rank < (p - 1)) {
        if (my_rank % 2 == 1) neighbor = (my_rank + 1) % p;
        else neighbor = (my_rank - 1) % p;
        MPI_Sendrecv(&myVal, 1, MPI_INT, neighbor, sTag, &recVal, 1, MPI_INT, neighbor, neighbor,
        MPI_COMM_WORLD, &status);
        if (abs(myVal - recVal) > diff) diff = abs(myVal - recVal);
    }
    MPI_Gather(&diff, 1, MPI_INT, d_val, 1, MPI_INT, 0, MPI_COMM_WORLD);
    if (my_rank == 0) {
        int dmax = d_val[0];
        for (int i = 1; i < p; i++) {
            printf("d_val[%d] = %d \n", i, d_val[i]);
            if (d_val[i] > dmax) dmax = d_val[i];
        }
        printf("\n Max = %d\n", dmax);
    }
    MPI_Finalize(); // Finalize the MPI environment.
}
```

$d\_val[1] = 4$   
 $d\_val[2] = 8$   
 $d\_val[3] = 16$   
 $d\_val[4] = 32$   
 $d\_val[5] = 64$   
 $d\_val[6] = 128$

$d\_val[7] = 128$   
 $Max = 128$

3 Attempt

1 Attempt

b) Can you interpret the meaning of the value of the variable dmax in the code above?

b) Can you interpret the meaning of the value of the variable dmax in the code above?

$2^{p-1}$ , where  $p = \text{Number of processors}$ .

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dmax value is giving value of  $2^{p-1}$  i.e.  $2^{8-1} = 128$

## Question # 2:

[6 marks, CLO # 3]

Assume these are the contents of sbuf in the processes P0, P1 and P2:

|    |   |   |   |    |    |   |
|----|---|---|---|----|----|---|
| P0 | 2 | 3 | 1 | 7  | 11 | 0 |
| P1 | 4 | 1 | 4 | 1  | 3  | 7 |
| P2 | 2 | 4 | 4 | 10 | 4  | 5 |

Show the contents of rbuf in the same processes after this MPI command:

`MPI_Scan(sbuf, rbuf, 6, MPI_INT, MPI_SUM, MPI_COMM_WORLD);`

Hint: (MPI\_Scan is used for prefix-sum).

Contents of rbuf:

|    |   |   |   |    |    |    |
|----|---|---|---|----|----|----|
| P0 | 2 | 3 | 1 | 7  | 11 | 0  |
| P1 | 6 | 4 | 5 | 8  | 14 | 7  |
| P2 | 8 | 8 | 9 | 18 | 18 | 12 |